# MD RAKIBUL ISLAM RAIHAN

*ML Systems & LLM Inference*

Email: raihanrakib.143@gmail.com
Phone: +86 177 2236 2949
GitHub: github.com/theraihanrakibb
LinkedIn: linkedin.com/in/theraihanrakib
Portfolio: theraihanrakibb.github.io

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## Education

**M.Eng. in Software Engineering** *Sep 2024 - Jul 2027 (expected)*
School of Software, Northwestern Polytechnical University, Xi'an, China

- Advisor: Prof. Zhang Peng
- GPA: 88/100 (Top 1%); Chinese Government Scholarship (2024-2027)

**B.Eng. in Computer Science & Technology** *Sep 2020 - Jul 2024*
School of Computer Science, Northwestern Polytechnical University, Xi'an, China

- Advisor: Dr. Yaxing Chen
- GPA: 85/100 (Top 1%); Presidential Scholarship; Best Undergraduate Thesis Award

## Research Interests

My research centers on efficient machine learning systems, with an emphasis on low-precision quantization for generative models and large-scale distributed LLM inference. I work on FP8/FP4/NVFP4 quantization and quantization-aware training, custom Triton/CUDA kernels, multimodal audio-visual generative models and deepfake detection, and KV-cache-centric serving architectures that trade storage for computation. A second line of work concerns privacy-preserving distributed data systems built on homomorphic encryption.

## Publications

First Author — MD Rakibul Islam Raihan, "NVFP4-DiT: Efficient 4-Bit Audio-Guided Video Diffusion Transformers for Low-Cost Video Generation" — *IEEE Transactions on Neural Networks and Learning Systems (TNNLS, Q1)*, manuscript under review.

## Research Experience

**Graduate Researcher | Sep 2024 – Present**
Audio, Speech and Language Processing Lab (ASLP@NPU)
School of Computer Science, Northwestern Polytechnical University
Advisor: Prof. Zhang Peng

Developed NVFP4-DiT, a 4-bit audio-guided video diffusion transformer with custom FP4-packed Triton kernels — 3.2× faster than FP16 (2.1× over naive FP4) and 4× memory reduction at competitive generation quality.

Implemented adaptive block-wise FP8/FP4/NVFP4 quantization and quantization-aware training with learnable per-head cross-modal scales; evaluated trade-offs on WebVid-10M, VGGSound, and UCF-101.

Built a vLLM-style serving scheduler (frame-level paged attention, dynamic frame batching) and benchmarked on H100 GPUs using PyTorch Profiler and Nsight to guide low-precision kernel design choices.

Conducted research on multimodal deepfake detection using visual, audio, and temporal consistency analysis.

M.Eng. Thesis: "Detecting Deepfake Video by a Multimodal Audio-Visual Framework with Temporal Inconsistencies."

**Undergraduate Research Assistant | Jan 2023 – Jul 2024**
NWPU Human-Cyber-Physical Fusion Intelligent Computing Innovation Center
School of Computer Science, Northwestern Polytechnical University
Advisor: Dr. Yaxing Chen

Designed and built a distributed confidential-query framework on Apache Spark + CKKS homomorphic encryption, validated on a 10-node cluster — achieving 95% privacy preservation and a 20% performance improvement via optimized analytics pipelines.

B.Eng. Thesis: "Design and Implementation of a Distributed Confidential Query Protocol for Apache Spark."

## Projects

**NVFP4-DiT**
A 4-bit audio-guided video diffusion transformer built with custom FP4-packed Triton kernels that are 3.2× faster than FP16 (2.1× over naive FP4) and cut memory 4× at competitive generation quality. Adds adaptive block-wise FP8/FP4/NVFP4 quantization-aware training with learnable per-head cross-modal scales, evaluated on WebVid-10M, VGGSound and UCF-101, plus a vLLM-style serving scheduler with frame-level paged attention benchmarked on H100 GPUs.

**AI Infrastructure Portfolio — github.com/theraihanrakibb/ai-infra-portfolio**
Ten Dockerized AI-infrastructure services - LLM gateway, model serving, GPU observability, KV-cache simulation, benchmarking, quantization, tracing and RAG - each with CI and a reproducible quickstart.

**Distributed Training Launcher — github.com/theraihanrakibb/train-launcher**
A PyTorch distributed CLI that launches and supervises multi-node training jobs with fault detection, automatic retry and checkpoint recovery.

## Industry Experience

**AI Infrastructure Engineer Intern** *Apr 2026 - Jun 2026*
InfiX.ai, Shenzhen, China
Production LLM serving on 4× A800-80GB and multi-node H200: 3-tier KV-cache offload over Mooncake RDMA (8.3× lower long-context TTFT, 55s→6.7s), 2+2 prefill/decode disaggregation (192→256 concurrent requests), and changes to SGLang cache-management and memory-allocation internals.

**Software Engineer Intern (AI Agent)** *Jun 2025 - Sep 2025*
Hong Kong Amraim Group Co., Ltd., Shenzhen, China
Delivered two end-to-end full-stack applications (React + RESTful services) and AI agents that automated marketing workflows, lifting team productivity by 25%.

**Electrical Software Engineer Intern** *Feb 2025 - Apr 2025*
Shaanxi Longong Intelligent Technology Co., Ltd., Xi'an, China
Real-time CODESYS PLC motor-control software over Modbus/CAN/Ethernet-IP; 15% faster system response to IEC 61131 industrial safety standards.

## Awards and Honors

- **Chinese Government Scholarship** 2024-2027
- **Presidential Scholarship, Northwestern Polytechnical University** 2020-2024
- **Wu Yajun Scholarship** 2024
- **Best Undergraduate Thesis Award** 2024
- **Outstanding Graduate, Northwestern Polytechnical University** 2024
- **Hackathon Winner, NWPU AI Challenge** 2023

## Services & Leadership

- **President, International Student Union (ISU)** Northwestern Polytechnical University

## Technical Skills

LLM Inference & Systems: SGLang, vLLM, KV Cache, PD Disaggregation, Mooncake/RDMA, Continuous Batching, RadixAttention
GPU & Performance: CUDA, CUDA C++, Triton, Nsight Systems, Nsight Compute, PyTorch Profiler, GEMM optimization, FP8/FP4/INT4
ML & Multimodal: PyTorch, Hugging Face Transformers, Diffusion/DiT, Multimodal Learning, OpenCV
Distributed Systems: PyTorch Distributed, Docker, Kubernetes, Linux, Prometheus, CI/CD
Programming: Python, C++, CUDA C++, Bash, SQL, Data Structures & Algorithms

## Certifications

Deep Learning Specialization · Claude Academy: Claude Code in Action · AI Agents Fundamentals